DC-OPS

On-Device Data Center Operations Assistant

Qualcomm x Meta ExecuTorch Hackathon — June 2026

Real-time AI-powered server rack inspection

Running entirely on Samsung Galaxy S25 Ultra — Snapdragon 8 Elite NPU

THE PROBLEM

🔒
Air-Gapped Environments
Data centers prohibit cloud connectivity in server rooms — no API calls possible

🕐
Slow Manual Inspection
Technicians walk aisles checking thousands of LEDs, ports, and cables by eye

🔐
Sensitive Infrastructure Data
Serial numbers, rack topology, and server status are classified information

📶
No Connectivity
Even connected DCs have dead zones between rack rows — WiFi is unreliable

THE SOLUTION

Point your phone at any server rack.

AI identifies every component in real-time.

All processing happens on-device — zero cloud.

16 Component Classes

Compute trays, ports, LEDs, cables, fans, drives, PSUs, DPUs, and more

RAG Knowledge Overlay

Tap any detection → specs, troubleshooting, LED meanings, maintenance steps

CPU vs NPU Toggle

Live comparison: XNNPACK (CPU) ↔ QNN HTP (Snapdragon NPU) with FPS/latency

WHY ON-DEVICE?

0ms

Cloud Latency

No round-trip — inference directly on Hexagon NPU

100%

Offline

Works in air-gapped rooms with zero connectivity

0 bytes

Data Leaked

Serial numbers and infra data never leave the device

4-5x

Power Efficient

NPU vs CPU — longer battery life for extended inspections

ARCHITECTURE

Camera
(CameraX 30fps)

RetinaNet
(INT8 Quantized)

QNN HTP
(Snapdragon NPU)

16 DC Classes
(Polygon Overlay)

3D NVL72 Rack
(Three.js Interactive)

RAG Knowledge Overlay: CLIP ViT-B-32 → FAISS (160KB) → Specs, Troubleshooting, LED Meanings

`Android App → ExecuTorch Runtime → QAIRT SDK → Snapdragon NPU`

TECH STACK

Detection (NPU)	RetinaNet-ResNet50-FPN	QNN HTP, INT8, 36MB
Detection (CPU)	YOLOv8n-seg v3	XNNPACK, FP32, 13MB
RAG Retrieval	CLIP ViT-B-32 + FAISS	160KB index, on-device
Runtime	ExecuTorch 1.4	QNN backend for SM8750
App	Kotlin + CameraX	ExecuTorch AAR (3.4MB)
Device	Galaxy S25 Ultra	Snapdragon 8 Elite, Hexagon v79
3D Visualization	Three.js NVL72 Model	Interactive rack explorer

TRAINING PIPELINE

PHASE 1 — Bootstrap

BrightData Web Unlocker scraped 300+ data-center images → auto-labeled with Grounding DINO + SAM (3,045 polygons)

PHASE 2 — 2,036 human-labeled images from 4 Roboflow datasets (CC BY 4.0) · used for final models

Server Vision	1,293 imgs	30 classes (CPU, DIMM, fans, PSUs, drives)
PC Ports	327 imgs	4 classes (HDMI, RJ45, USB-A, USB-C)
Ports (ym)	138 imgs	7 classes (DP, ethernet, power, USB)
Server Detection	54 imgs	9 classes (SFP, XFP, patch panels)

50+ source classes → mapped to 16 DC-Ops classes · Augmentations: Moiré, screen glare, brightness, color tint, scanlines

⚡ ***Powered by BrightData — web data infrastructure for the bootstrap dataset***

MODEL PERFORMANCE

0.749
mAP50 (box)
YOLOv8n-seg v3

0.85
Final Loss
RetinaNet (moiré aug)

6.4ms
Inference
Per frame on T4 GPU

compute tray	0.924	server rack	0.683
network port	0.912	drive bay	0.912
power shelf	0.832	DPU	0.808
LED indicator	0.808	cooling manifold	0.833

LIVE DEMO

0:00 — **Point phone at server rack / mini PC**

Real-time colored polygon overlays appear

1:00 — **Tap detected component**

RAG info panel: specs, troubleshooting, LED meanings

2:00 — **Toggle CPU ↔ NPU**

Live FPS/latency comparison on screen

3:00 — **Enable airplane mode**

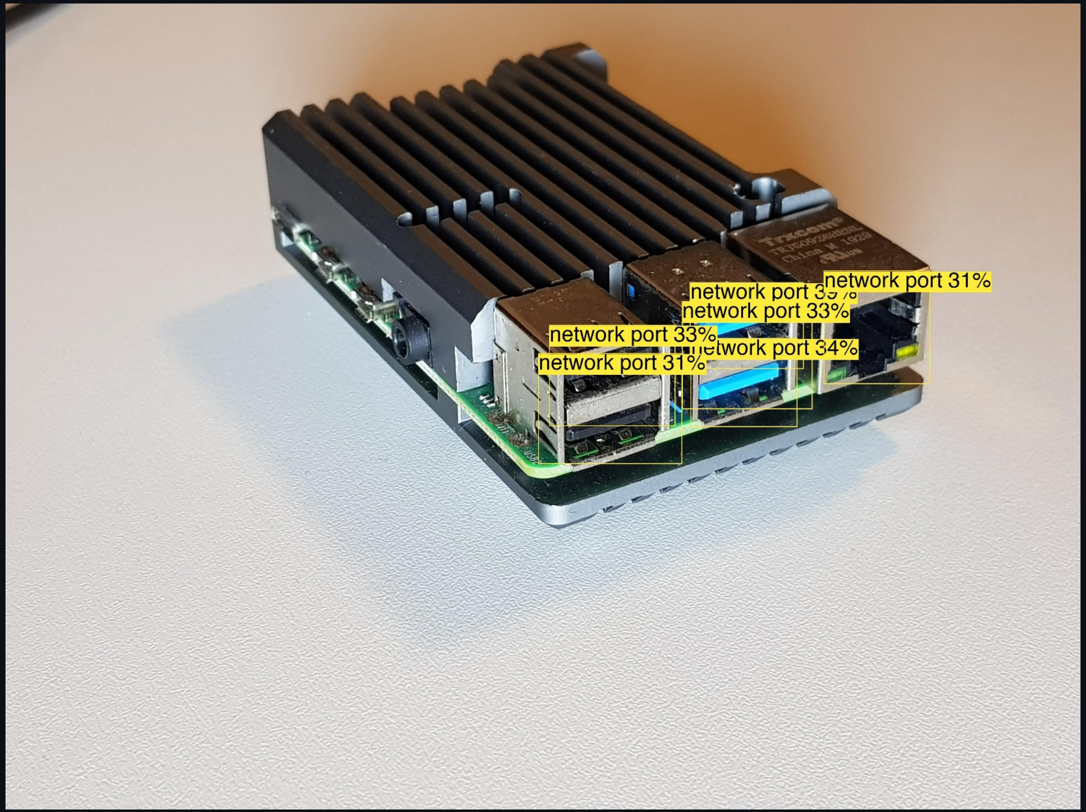
Everything still works — fully offline

4:00 — **3D rack explorer**

Interactive NVL72 model with component details

RETINANET IN ACTION

Real detections on a Raspberry Pi & USB hub — mirrors our physical demo prop



All USB + RJ45 ports detected as 'network port' — fully on-device

RAG KNOWLEDGE OVERLAY

Detect → Embed → Retrieve → Display

```
COMPUTE TRAY
  92%

⚙  2x Grace CPU, 4x B200 GPU, 192GB HBM3e

🟢  Green LED = healthy and operational
🟡  Amber LED = check BMC log, verify coolant
🔴  Red LED = critical hardware failure

🔧  Maintenance: check cold plate contact,
     verify NVLink connector seating
```

- 80 knowledge chunks
- CLIP ViT-B-32 embeddings
- FAISS vector index (160KB)
- NVL72-specific documentation
- All on-device, zero cloud

CPU vs NPU COMPARISON

Backend	XNNPACK (CPU)	QNN HTP (NPU)
Model	YOLOv8n-seg	**RetinaNet**
Size	13 MB	**36 MB**
Quantization	FP32	**INT8**
Hardware	ARM Kryo cores	**Hexagon HTP v79**
Latency †	~15-30 ms	**~3-5 ms**
Throughput †	~30-60 FPS	**~100+ FPS**
Power †	Baseline	**4-5x efficient**

† Estimated from Qualcomm QNN HTP benchmarks — not yet measured on our specific model. Measured FPS shown live in-app.

WHAT'S NEXT

Fine-grained Port Classification

Split 'network port' into HDMI, USB-A, USB-C, RJ45, SFP individually

LED State Detection

Classify LED colors (green/amber/red) and blink patterns for health scoring

On-Device LLM Integration

LLaMA 3.2 1B via ExecuTorch for natural language troubleshooting Q&A

Audit Log + Reporting

SQLite database of all inspections, exportable via secure channel

Multi-Rack Mapping

Combine depth estimation + detection for spatial rack layout understanding

DC-OPS

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GitHub	`github.com/abhijitbetigeri/DC-Ops`
HuggingFace	`huggingface.co/datasets/abhijitbetigeri/dc-ops-dataset`
Models	`RetinaNet QNN HTP (.pte) + YOLOv8n-seg (.pte) + RAG index`

Built with ExecuTorch + Qualcomm QNN HTP + Snapdragon 8 Elite

Thank you!